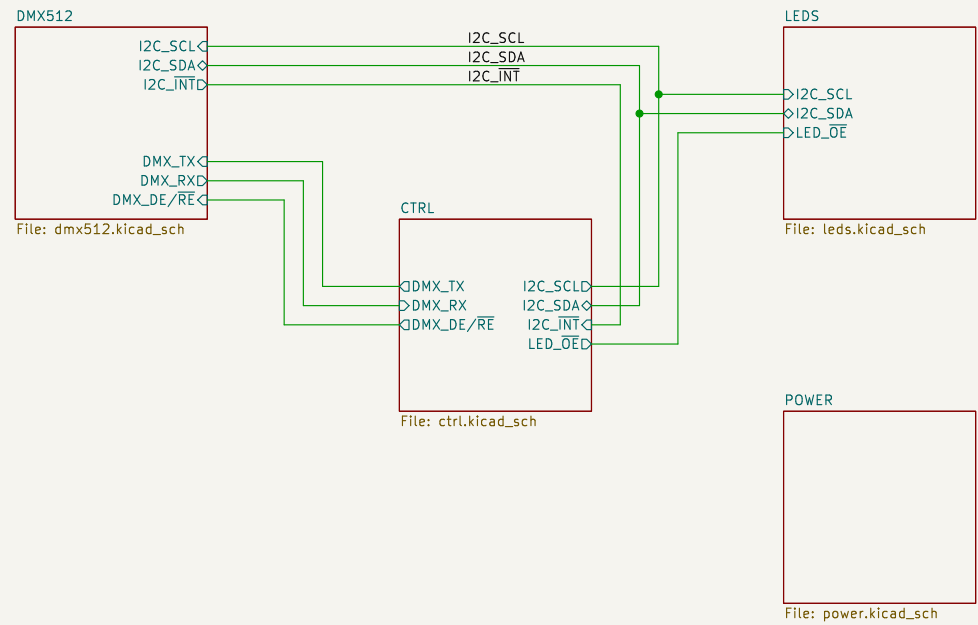


# LED Beacon



- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole

|                            |                  |         |
|----------------------------|------------------|---------|
| <b>Wires &amp; Bits</b>    |                  |         |
| Sheet: /                   |                  |         |
| File: emf_beacon.kicad_sch |                  |         |
| <b>Title: LED Beacon</b>   |                  |         |
| Size: A4                   | Date: 2026-06-05 | Rev: 1  |
| KiCad E.D.A. 9.0.9         |                  | Id: 1/5 |



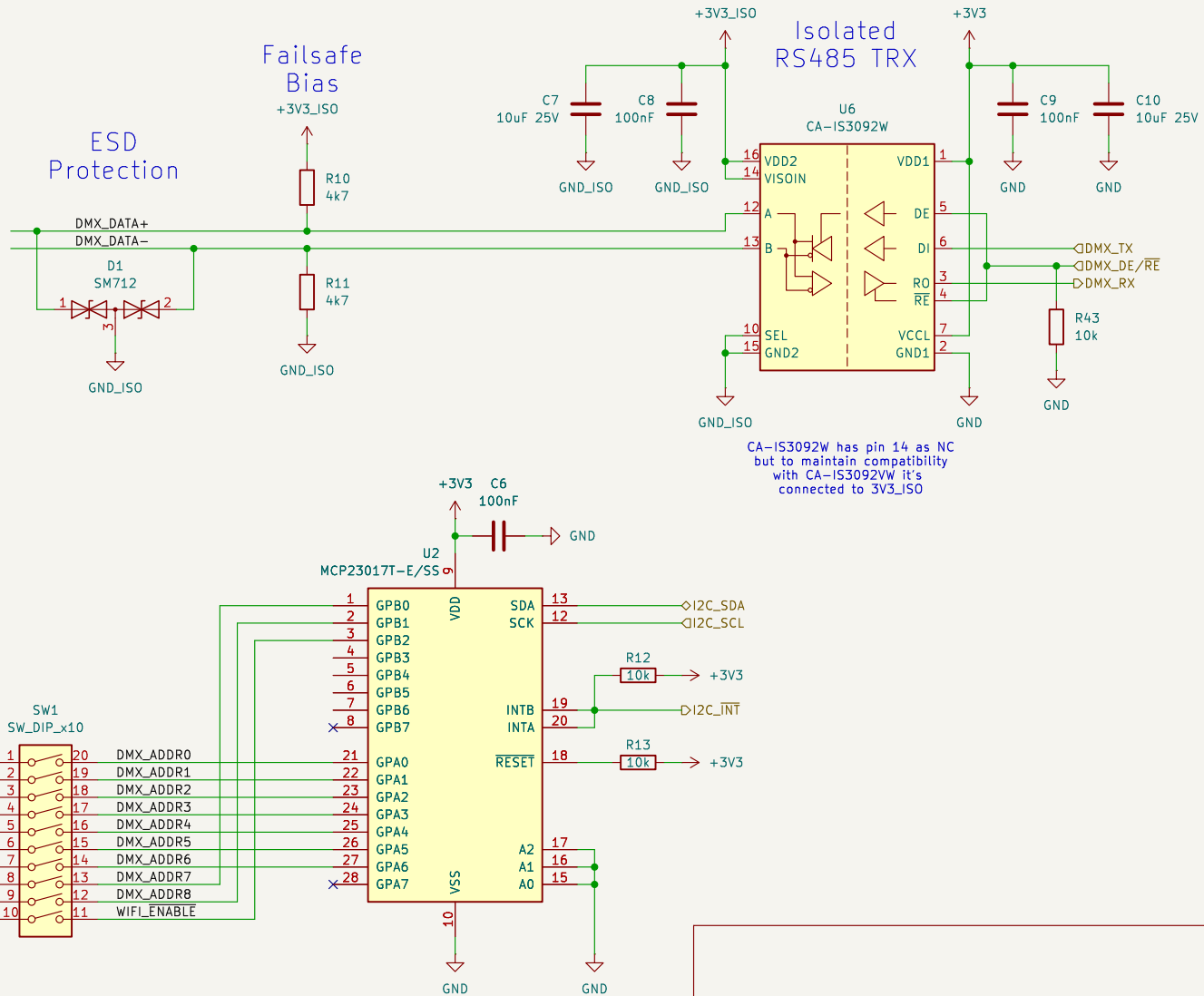
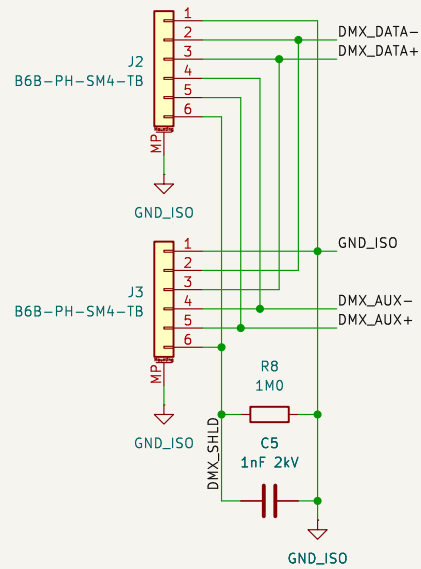
# DMX512

Failsafe Bias  
+3V3\_ISO

ESD Protection

Isolated RS485 TRX

To XLRs



CA-IS3092W has pin 14 as NC but to maintain compatibility with CA-IS3092VW it's connected to 3V3\_ISO

I<sup>2</sup>C Addr: 0x20  
MCP23017 has a fault with GPA7 & GPB7.  
The can corrupt an I<sup>2</sup>C transaction if they change state during it.

## Wires & Bits

Sheet: /DMX512/  
File: dmx512.kicad\_sch

## Title: DMX512

Size: A4 Date: 2026-06-05  
KiCad E.D.A. 9.0.9

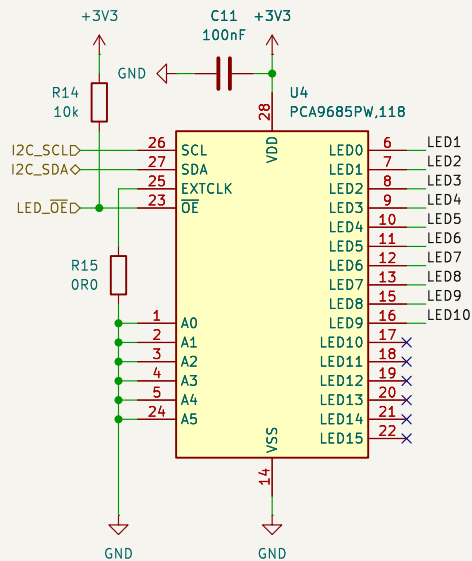
Rev: 1  
Id: 3/5

# LED Driver

## LED Enable

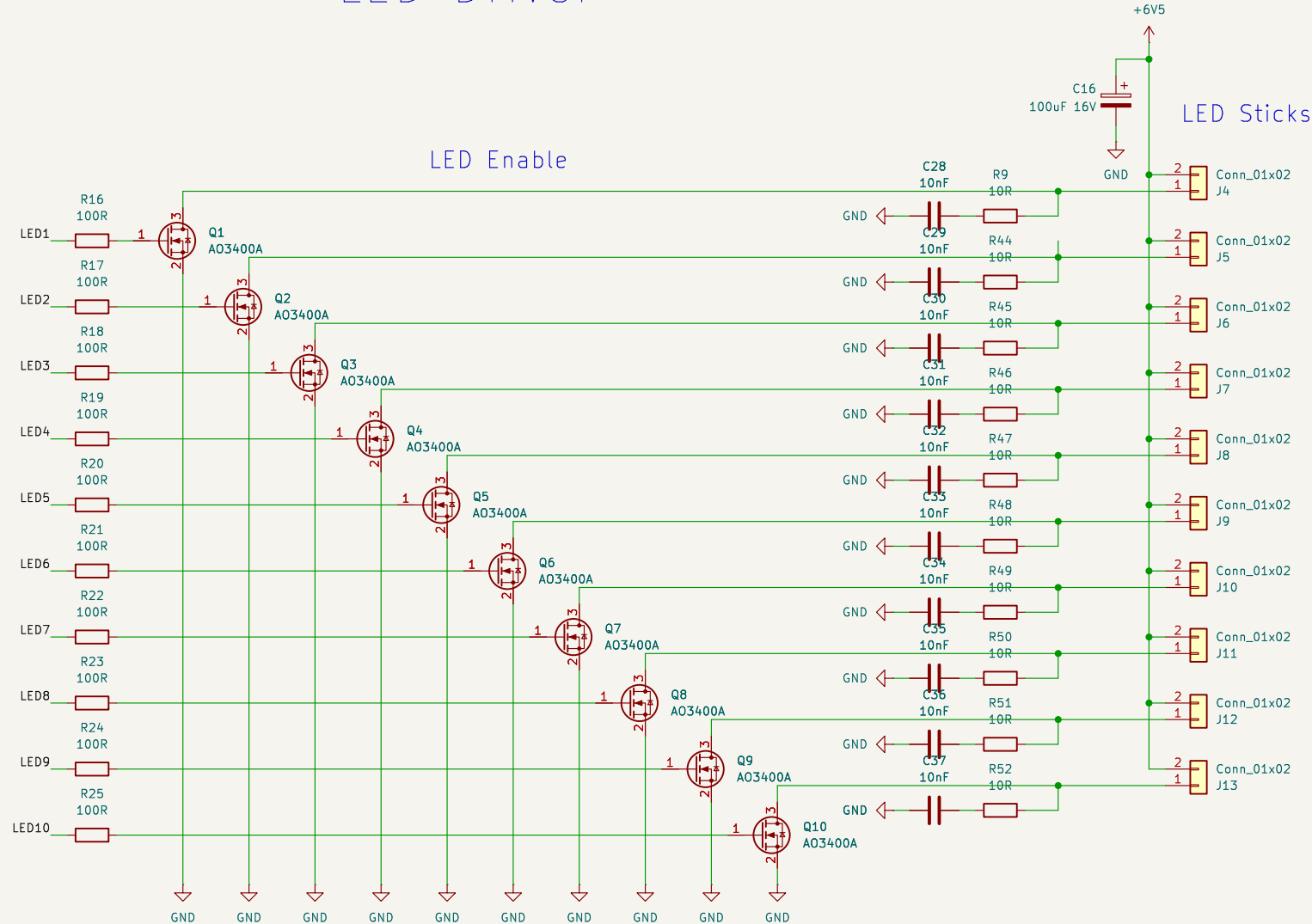
## LED Sticks

### I2C → PWM



I2C Addr: 0x40

A string of 6x LEDs is 6.4Vf @ 0.3A



### Wires & Bits

Sheet: /LEDS/  
File: leds.kicad\_sch

### Title: LEDs

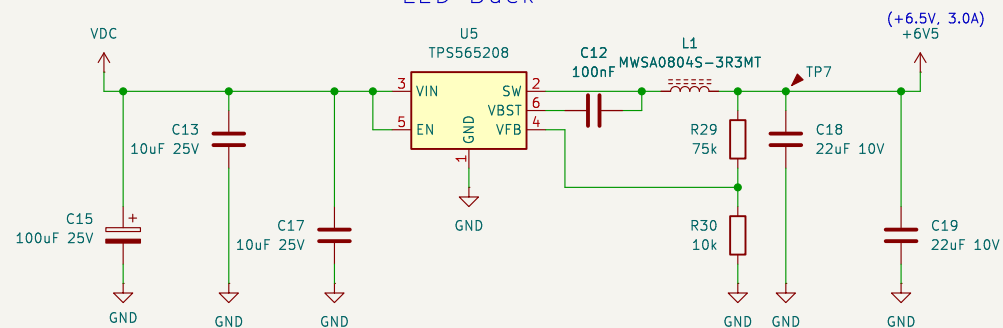
Size: A4  
KiCad E.D.A. 9.0.9

Date: 2026-06-05

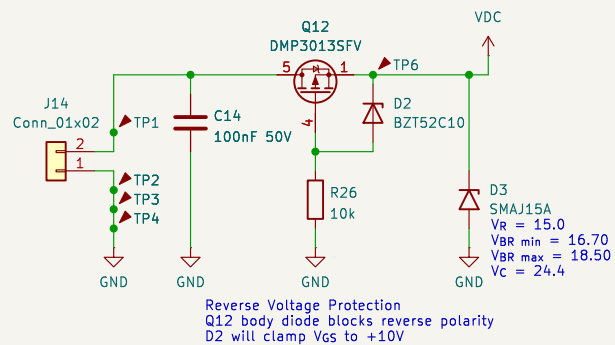
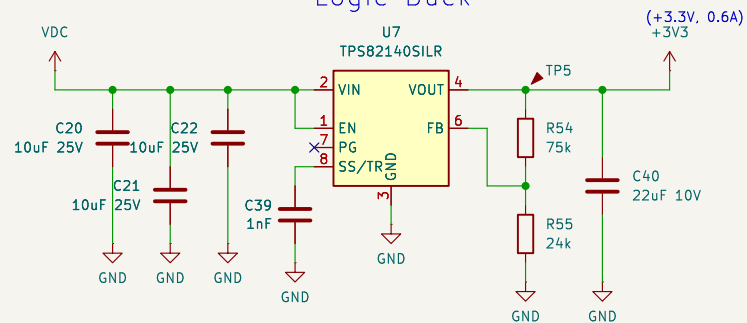
Rev: 1  
Id: 4/5

# Power Supplies

## LED Buck



## Logic Buck



EMC ferrites /CMC?

### Wires & Bits

Sheet: /POWER/  
File: power.kicad\_sch

### Title: Power

Size: A4  
Date: 2026-06-05  
KiCad E.D.A. 9.0.9

Rev: 1  
Id: 5/5